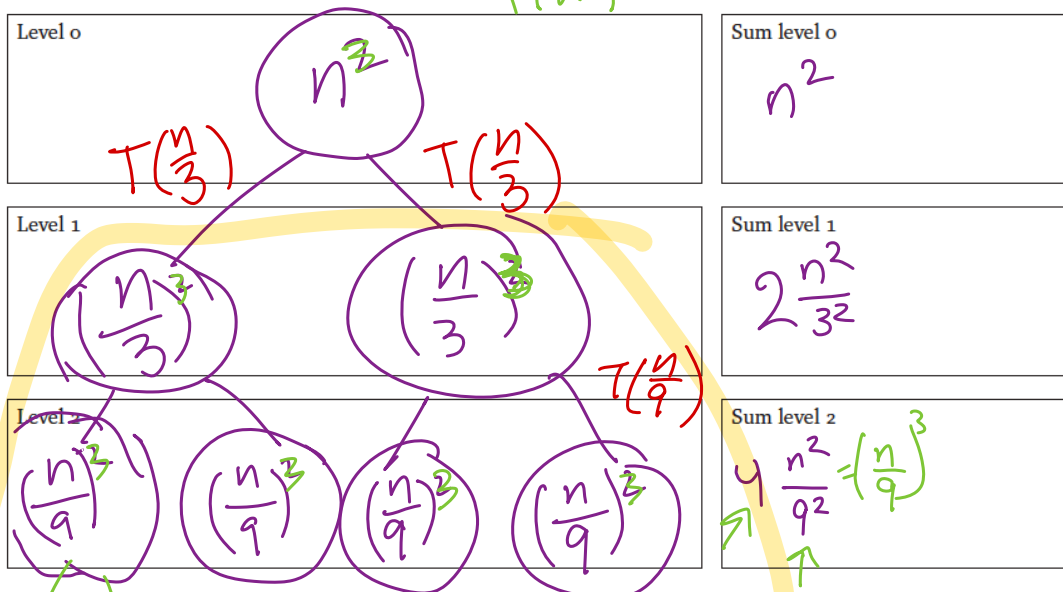


Consider the recurrence $T(n) = 2T(n/3) + n^2, T(1) = 1$.

2. (2 points) In the left boxes, draw the first three levels of this recursion tree. In the boxes next to them, write the total work on that level.



3. (2 points) What is the work per level in terms of level l ? You should provide a function of n and l here.

$$2^l \cdot \left(\frac{n}{3^l}\right)^2 = 2^l \left(\frac{n^2}{3^{2l}}\right) = 2^l \cdot \frac{n^2}{(3^2)^l} = 2^l \cdot \frac{n^2}{9^l}$$

$$= \left(\frac{2}{9}\right)^l n^2$$

$$\rightarrow \left(\frac{10}{9}\right)^l n^2$$

4. (2 points) Is the work per level (circle one)

- ☒ increasing (work is dominated by leaves) $\rightarrow$
☒ decreasing (work is dominated by root) $\rightarrow$
☐ the same (work is the same at every level) $\leftarrow$

5. (1 points) What is the total number of levels in the tree? (fill in the base and the argument of the logarithm)

number of levels = $\log_3 (n)$

6. (2 points) What is the overall runtime of the recurrence?

$$n^2$$

- given rec alg, write recurrence
- given a recurrence, find its Closed-Form Solution

now: writing rec. algs

→ transform input into smaller instances of the same problem

L half
R half

→ recurse

MS L half
MS R half

→ post processing

merge

MysterySort(array A of length n):

if $n \leq 1$:
return A

EX
XELLOW

else: swap min elt of A into 1st position

return MysterySort(rest of A)
first elt of A

$$T(n) = \begin{cases} T(n-1) + n & \text{if } n > 1 \\ \theta(1) & \text{if } n \leq 1 \end{cases}$$

